UNIVERSITY OF OSLO

DEXPI 2.0 modelling of CO_2 Sequestration

OPC Material and Resource Flow

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4th May 2026



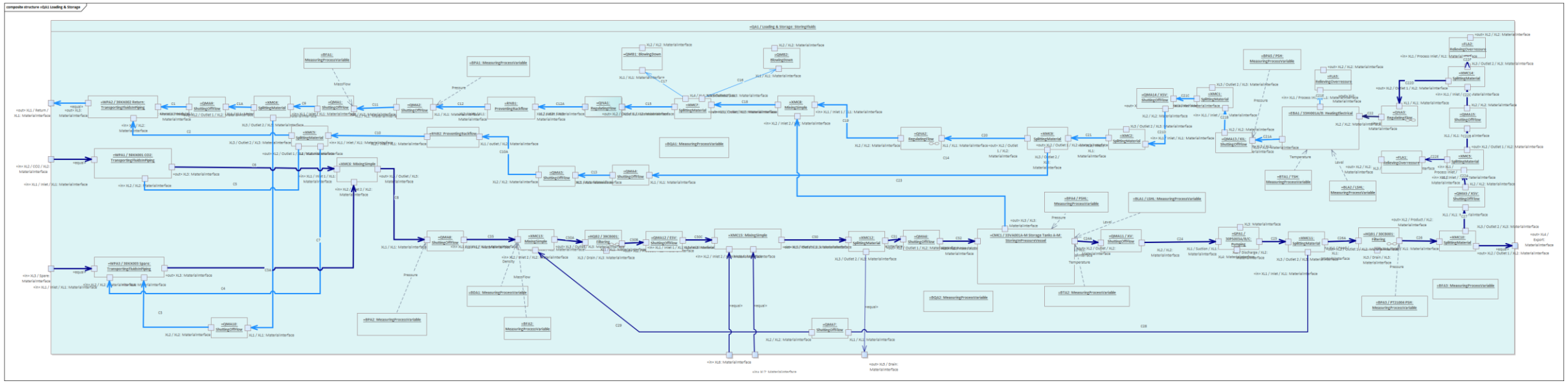
The Problem

- Represent process and safety information for a CO_2 sequestration plant.
  - Structure
  - Process Flow Diagram level of detail
  - Supervisory control and process information.
  - Process simulation information
- Contextualize sensor data
  - Support validation and checking against design specifications and alarm limits.

The Solution

- Model the process using DEXPI 2 process package.
- Implemented at present in two formats:
  - An Information Modelling Framework model
    - Configured in Enterprise Architect SysML 1.5
    - Using *types* defined by DEXPI 2.
  - An XML file containing the DEXPI representation of:
    - The Process Flow Diagram
    - Stream and analysis properties
    - Measured variables and supervisory control
      - As shown on the process control mimic.

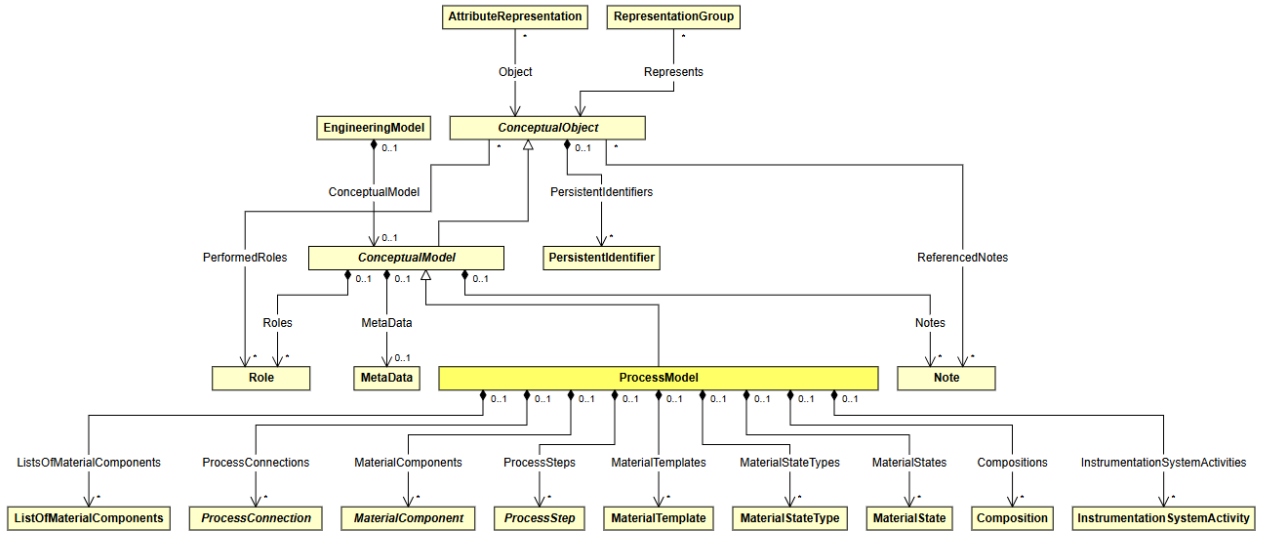
Process Model in SysML / IMF format



DEXPI XML

```
<Model name="CO2Injection" uri="" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:noNamespaceSchemaLocation="DEXPI_XML_Schema.xsd">
 <Import prefix="Core" source="https://data.dexpi.org/models/2.0.0/Core.xml"/>
 <Import prefix="MetaData" source="https://data.dexpi.org/models/2.0.0/MetaData.xml"/>
 <Import prefix="Process" source="https://data.dexpi.org/models/2.0.0/Process.xml"/>
 <Object type="Core/EngineeringModel">
 <Data property="OriginatingSystemName">
   <String>XML Spy</String>
 </Data>
 <Data property="OriginatingSystemVendorName">
   <String>University of Oslo</String>
 </Data>
 <Data property="OriginatingSystemVersion">
   <String>2.0</String>
 </Data>
 <Components property="ConceptualModel">
 <Object type="Process/ProcessModel" id="ProcessModel1">...</Object>
 </Components>
 </Object>

 <!-- The example is put into an InstrumentLoopFunction -->
 </Model>
```



Procedure

- Define stream and analysis structure
- Define process steps
  - Blocks
  - Ports
- Define instrumentation functions
- Define conections

Declare Chemical Components

```
<Components property="MaterialComponents">
    <Object type="Process/Process.PureMaterialComponent" id="CO2">
        <Data property="Identifier"><String>CO2</String></Data>
        <Data property="Label"><String>CO2</String></Data>
        <Data property="Description">
            <AggregatedDataValue type="Core/DataTypes.MultiLanguageString">
                <Data property="SingleLanguageStrings">
                    <AggregatedDataValue type="Core/DataTypes.SingleLanguageString">
                        <Data property="Language"><String>en</String></Data>
                        <Data property="Value"><String>Carbon Dioxide</String></Data>
                    </AggregatedDataValue>
                </Data>
            </AggregatedDataValue>
        </Data>
        <Data property="ChEBI_Identifier"><String>CHEBI:16526</String></Data>
        <Data property="IUPAC_Identifier"><String>carbon dioxide</String></Data>
    </Object>

    <Object type="Process/Process.PureMaterialComponent" id="H2O">
        <Data property="Identifier"><String>H2O</String></Data>
        <Data property="Label"><String>H2O</String></Data>
        <Data property="Description">
            <AggregatedDataValue type="Core/DataTypes.MultiLanguageString">
                <Data property="SingleLanguageStrings">
                    <AggregatedDataValue type="Core/DataTypes.SingleLanguageString">
                        <Data property="Language"><String>en</String></Data>
                        <Data property="Value"><String>Water</String></Data>
                    </AggregatedDataValue>
                </Data>
            </AggregatedDataValue>
        </Data>
        <Data property="ChEBI_Identifier"><String>CHEBI:15377</String></Data>
    </Object>
```

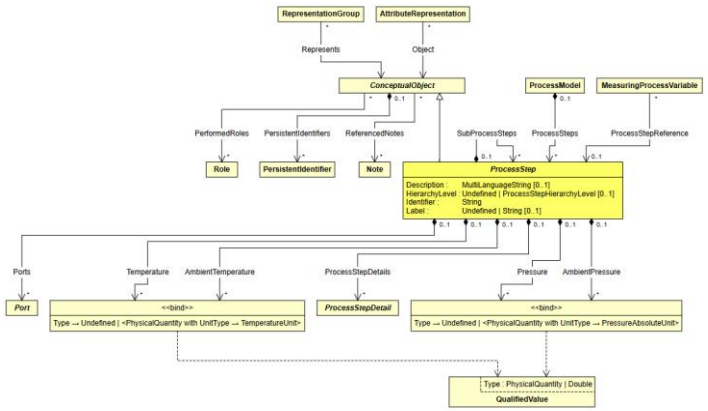
Create List of Components

```
<Components property="ListsOfMaterialComponents">
    <Object type="Process/Process.ListOfMaterialComponents" id="v">
        <References property="Component" objects="#CO2"/>
        <References property="Component" objects="#Trimethylamine"/>
        <References property="Component" objects="#DiethylEther"/>
        <References property="Component" objects="#Argon"/>
        <References property="Component" objects="#BTEX"/>
        <References property="Component" objects="#Ethylamine"/>
        <References property="Component" objects="#MEG"/>
        <References property="Component" objects="#Aliphatics"/>
        <References property="Component" objects="#MethylAcetate"/>
        <References property="Component" objects="#Butane"/>
        <References property="Component" objects="#Trimethlybenzene"/>
        <References property="Component" objects="#Ethylbenzene"/>
        <References property="Component" objects="#Toluene"/>
        <References property="Component" objects="#Benzene"/>
        <References property="Component" objects="#Pentane"/>
        <References property="Component" objects="#Butanol"/>
        <References property="Component" objects="#Propanol"/>
        <References property="Component" objects="#Propane"/>
        <References property="Component" objects="#Ethanol"/>
        <References property="Component" objects="#Ethane"/>
        <References property="Component" objects="#Acetaldehyde"/>
        <References property="Component" objects="#Acetonitrile"/>
        <References property="Component" objects="#Acetone"/>
        <References property="Component" objects="#Methylamine"/>
        <References property="Component" objects="#Methanol"/>
        <References property="Component" objects="#Methane"/>
        <References property="Component" objects="#CO"/>
        <References property="Component" objects="#Hydrogen"/>
        <References property="Component" objects="#Monoethanolamine"/>
        <References property="Component" objects="#H2O"/>
        <References property="Component" objects="#H2S"/>
        <References property="Component" objects="#Formaldehyde"/>
        <References property="Component" objects="#HCN"/>
        <References property="Component" objects="#Ethylene"/>
        <References property="Component" objects="#N2"/>
        <References property="Component" objects="#N2O"/>
        <References property="Component" objects="#NH3"/>
        <References property="Component" objects="#NO"/>
        <References property="Component" objects="#NO2"/>
        <References property="Component" objects="#O2"/>
        <References property="Component" objects="#SO2"/>
        <References property="Component" objects="#TotalSulphur"/>
        <References property="Component" objects="#TotalVOC"/>
        <References property="Component" objects="#mXylene"/>
        <References property="Component" objects="#oXylene"/>
        <References property="Component" objects="#pXylene"/>
    </Object>
</Components>
```

Material Template

```
<Components property="MaterialTemplates">
    <Object type="Process/Process.MaterialTempate" id="CO2AnalysisTemplate">
        <References property="ListOfComponents" objects="#CO2AnalysisComponents"/>
        <Data property="Identifier"><String>CO2Stream</String></Data>
        <Data property="Label"><String>CO2 Stream</String></Data>
        <Data property="NumberOfMaterialComponents"><Integer>46</Integer></Data>
        <Data property="NumberOfPhases"><Integer>2</Integer></Data>
        <Data property="PhaseLabel"><String>Liquid</String></Data>
        <Data property="PhaseLabel"><String>Vapour</String></Data>
    </Object>
</Components>
<!--
```

Process Step




```
<Components property="ProcessSteps">
    <Object type="Process/Process.TransportingFluids" id="Ship">
        <Data property="Identifier"><String>=JD1</String></Data>
        <Data property="Label"><String>=JD1</String></Data>
        <Data property="Description"><AggregatedDataValue type="Core/DataTypes.MultiLanguageString">
        <Data property="SingleLanguageStrings">
            <AggregatedDataValue type="Core/DataTypes.SingleLanguageString">
                <Data property="Language"><String>en</String></Data>
                <Data property="Value"><String>Liquid CO2 Delivery Ship</String></Data>
            </AggregatedDataValue>
            <AggregatedDataValue type="Core/DataTypes.SingleLanguageString">
                <Data property="Language"><String>no</String></Data>
                <Data property="Value"><String>Flytende CO2 Leveringsskip</String></Data>
                </AggregatedDataValue>
            </Data></AggregatedDataValue>
        </Data>
        <Components property="PersistentIdentifiers">
            <Object type="Core/PersistentIdentifier">
                <Data property="Context"><String>ISO/IEC81346</String></Data>
                <Data property="Value"><String>=JD1</String></Data>
            </Object></Components>
        <Components property="Ports">
            <Object type="Process/Process.MaterialPort" id="Ship_XL1">
                <Components property="PersistentIdentifiers">
                    <Object type="Core/PersistentIdentifier">
                        <Data property="Context"><String>ISO/IEC81346</String></Data>
                        <Data property="Value"><String>=JD1:XL1</String></Data>
                    </Object></Components>
                <References property="ConnectorReference" objects="#Stream1"></References>
                <References property="MaterialTemplateReference" objects="#CO2AnalysisTemplate"/>
                <Data property="NominalDirection"><DataReference data="Process/Enumerations.PortDirection.Outlet"/></Data>
            </Object>
            <Object type="Process/Process.MaterialPort" id="Ship_XL2">
                <Components property="PersistentIdentifiers">
                    <Object type="Core/PersistentIdentifier">
                        <Data property="Context"><String>ISO/IEC81346</String></Data>
                        <Data property="Value"><String>=JD1:XL2</String></Data>
                    </Object></Components>
                <References property="ConnectorReference" objects="#Stream2"></References>
                <Data property="NominalDirection"><DataReference data="Process/Enumerations.PortDirection.Inlet"/></Data>
            </Object>
                <Object type="Process/Process.MaterialPort" id="Ship_XL3">
                <Components property="PersistentIdentifiers">
                    <Object type="Core/PersistentIdentifier">
                        <Data property="Context"><String>ISO/IEC81346</String></Data>
                        <Data property="Value"><String>=JD1:XL3</String></Data>
                    </Object></Components>
                <References property="ConnectorReference" objects="#Stream3"></References>
                <Data property="NominalDirection"><DataReference data="Process/Enumerations.PortDirection.Outlet"/></Data>
            </Object>
        </Components>
    </Object>
```

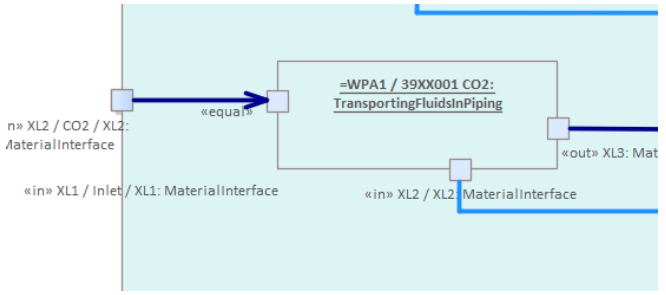
Identification

Reference designation

Ports

Embedding Sub-processes

```
<Object type="Process/Process.StoringFluids" id="StoringFluids1">
    <Data property="Identifier"><String>=QA1</String></Data>
    <Data property="Label"><String>Ship</String></Data>
    <Data property="Description"><AggregatedDataValue type="Core/DataTypes.MultiLanguageString">
    <Data property="SingleLanguageStrings">
        <AggregatedDataValue type="Core/DataTypes.SingleLanguageString">...</AggregatedDataValue>
        <AggregatedDataValue type="Core/DataTypes.SingleLanguageString">...</AggregatedDataValue>
        </Data></AggregatedDataValue>
    </Data>
    <Components property="PersistentIdentifiers">...</Components>
    <Components property="SubProcessSteps">
        <Object type="Process/TransportingFluidsInPiping" id="TransportingFluidsInPiping1">
            <Data property="Identifier"><String>=QA1.WPA1</String></Data>
            <Data property="Label"><String>39XX001</String></Data>
            <Data property="Description"><AggregatedDataValue type="Core/DataTypes.MultiLanguageString">
                <Data property="SingleLanguageStrings">
                    <AggregatedDataValue type="Core/DataTypes.SingleLanguageString">
                        <Data property="Language"><String>en</String></Data>
                        <Data property="Value"><String>Loading Arm 1</String></Data>
                    </AggregatedDataValue>
                </Data></AggregatedDataValue>
            </Data>
            <Components property="PersistentIdentifiers">
                <Object type="Core/PersistentIdentifier">
                    <Data property="Context"><String>ISO/IEC81346</String></Data>
                    <Data property="Value"><String>=QA1.WPA1</String></Data>
                </Object>
            </Components>
            <Components property="Ports">
                <Object type="Process/Process.MaterialPort" id="TransportingFluidsInPiping1_XL1">
                    <Components property="PersistentIdentifiers">
                        <Object type="Core/PersistentIdentifier">
                            <Data property="Context"><String>ISO/IEC81346</String></Data>
                            <Data property="Value"><String>=QA1.WPA1:XL1</String></Data>
                        </Object>
                    </Components>
                    <References property="SuperReference" objects="#StoringFluids1_XL2"/>
                    <Data property="NominalDirection">
                        <DataReference data="Process/Enumerations.PortDirection.Inlet"/>
                    </Data>
                </Object>
                <Object type="Process/Process.MaterialPort" id="TransportingFluidsInPiping1_XL2">
                    <Components property="PersistentIdentifiers">
                        <Object type="Core/PersistentIdentifier">
                            <Data property="Context"><String>ISO/IEC81346</String></Data>
                            <Data property="Value"><String>=QA1.WPA1:XL2</String></Data>
                        </Object>
                    </Components>
                    <References property="ConnectorReference" objects="#C5"></References>
                    <Data property="NominalDirection">
                        <DataReference data="Process/Enumerations.PortDirection.Inlet"/>
                    </Data>
                </Object>
```



Sub-processes

Super-reference for ports

Connectors with locally unique identifier

Instrumentation and Control

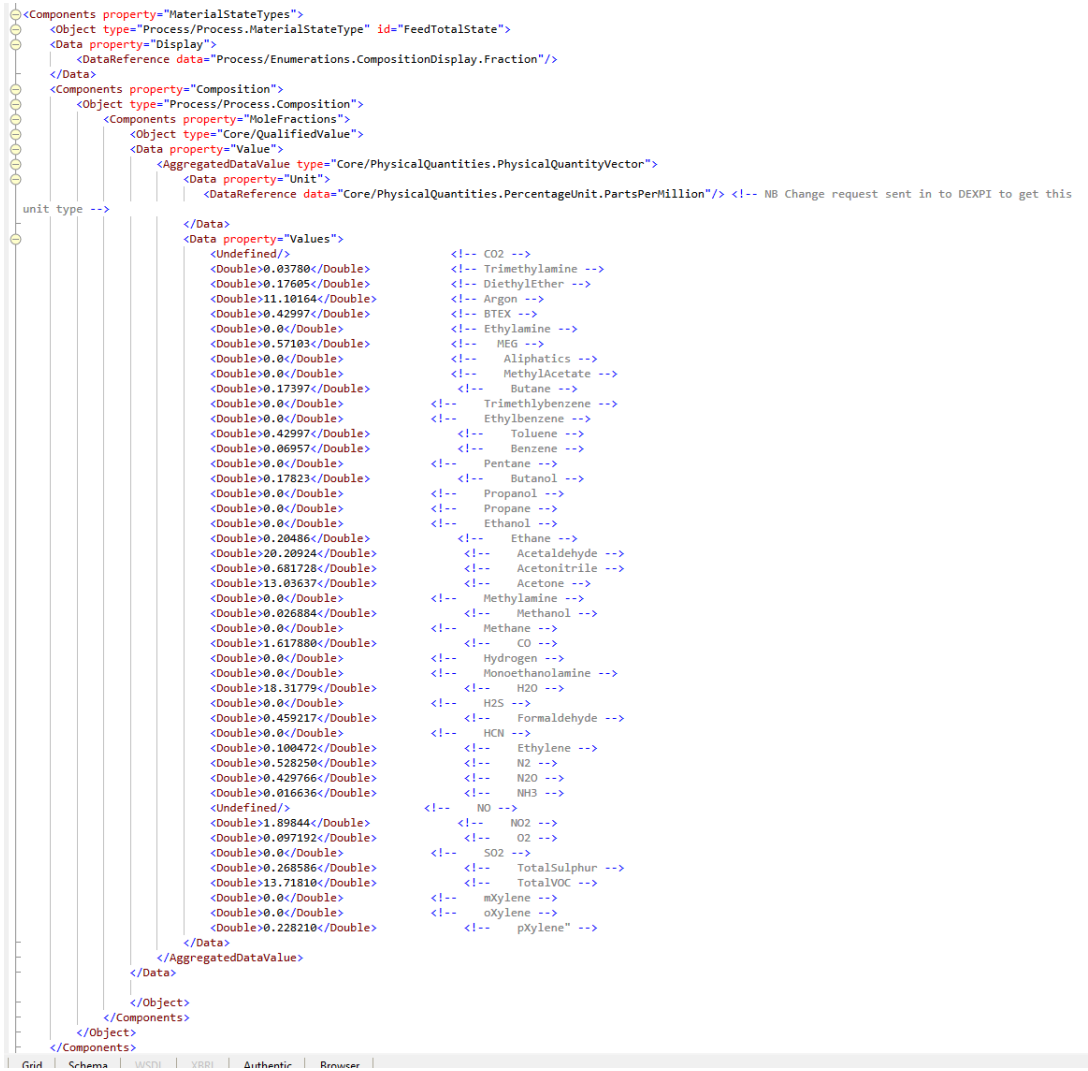
```
<Components property="InstrumentationSystemActivities">
    <Object type="Process/InstrumentationSystemActivity" id="PSH">
    <Components property="InstrumentationActivities">
        <Object type="Process/MeasuringProcessVariable" id="PSHB30">
            <Data property="Identifier"><String>=JA1.BPA1</String></Data>
            <Data property="Label"><String>=JA1.BPA1</String></Data>
            <Components property="PersistentIdentifiers">
                <Object type="Core/PersistentIdentifier">
                    <Data property="Context"><String>ISO/IEC81346</String></Data>
                    <Data property="Value"><String>=JA1.BPA1</String></Data>
                </Object>
            </Components>
            <References property="ConnectionReference" objects="#B30"/>
            <References property="MeasuredVariableReference" objects="#PressureB30"/>
        </Object>

        <Object type="Process/MeasuringProcessVariable" id="PSHL33VA001">
            <Data property="Identifier"><String>=QA1.BPA4</String></Data>
            <Data property="Label"><String>=QA1.BPA4</String></Data>
            <Components property="PersistentIdentifiers">
                <Object type="Core/PersistentIdentifier">
                    <Data property="Context"><String>ISO/IEC81346</String></Data>
                    <Data property="Value"><String>=QA1.BPA4</String></Data>
                </Object>
            </Components>
            <References property="ProcessStepReference" objects="#StoringInPressureVessel1"/>
            <References property="MeasuredVariableReference" objects="#StoringInPressureVessel1Pressure"/>
        </Object>
    </Components>
    </Object>
```

Process Connections

```
<Object type="Process/Process.Stream" id="C8">
    <Components property="Pressure">
        <Object type="Core/QualifiedValue" id="PressureC8">
            <Data property="Value">
                <AggregatedDataValue type="Core/PhysicalQuantities.PhysicalQuantity">
                    <Data property="Unit">
                        <DataReference data="Core/PhysicalQuantities.PressureAbsoluteUnit.Bar"/>
                    </Data>
                    <Data property="Value"><Double>5.0</Double></Data>
                </AggregatedDataValue>
            </Data>
        </Object>
    </Components>
    <References property="Source" objects="#MixingSimple6_XL3"/>
    <References property="Target" objects="#ShuttingOffFlow8_XL1"/>
</Object>
```

Analysis and Stream Tables



```
<Components property="MaterialStateTypes">
    <Object type="Process/Process.MaterialStateType" id="FeedTotalState">
    <Data property="Display">
        <DataReference data="Process/Enumerations.CompositionDisplay.Fraction"/>
    </Data>
    <Components property="Composition">
        <Object type="Process/Process.Composition">
            <Components property="MoleFractions">
                <Object type="Core/QualifiedValue">
                <Data property="Value">
                    <AggregatedDataValue type="Core/PhysicalQuantities.PhysicalQuantityVector">
                        <Data property="Unit">
                          <DataReference data="Core/PhysicalQuantities.PercentageUnit.PartsPerMillion"/> <!-- NB Change request sent in to DEXPI to get this
 unit type -->
                        </Data>
                        <Data property="Values">
                            <Undefined/>                    <!-- CO2 -->
                            <Double>0.03780</Double>        <!-- Trimethylamine -->
                            <Double>0.17605</Double>        <!-- DiethylEther -->
                            <Double>11.10164</Double>       <!-- Argon -->
                            <Double>0.42997</Double>        <!-- BTEX -->
                            <Double>0.0</Double>            <!-- Ethylamine -->
                            <Double>0.57103</Double>        <!--   MEG -->
                            <Double>0.0</Double>            <!--    Aliphatics -->
                            <Double>0.0</Double>            <!--    MethylAcetate -->
                            <Double>0.17397</Double>         <!--    Butane -->
                            <Double>0.0</Double>          <!--    Trimethlybenzene -->
                            <Double>0.0</Double>          <!--    Ethylbenzene -->
                            <Double>0.42997</Double>          <!--    Toluene -->
                            <Double>0.06957</Double>          <!--    Benzene -->
                            <Double>0.0</Double>          <!--    Pentane -->
                            <Double>0.17823</Double>          <!--    Butanol -->
                            <Double>0.0</Double>          <!--    Propanol -->
                            <Double>0.0</Double>          <!--    Propane -->
                            <Double>0.0</Double>          <!--    Ethanol -->
                            <Double>0.20486</Double>           <!--    Ethane -->
                            <Double>20.20924</Double>           <!--    Acetaldehyde -->
                            <Double>0.681728</Double>           <!--    Acetonitrile -->
                            <Double>13.03637</Double>           <!--    Acetone -->
                            <Double>0.0</Double>          <!--    Methylamine -->
                            <Double>0.026884</Double>          <!--    Methanol -->
                            <Double>0.0</Double>          <!--    Methane -->
                            <Double>1.617880</Double>          <!--    CO -->
                            <Double>0.0</Double>          <!--    Hydrogen -->
                            <Double>0.0</Double>          <!--    Monoethanolamine -->
                            <Double>18.31779</Double>          <!--    H2O -->
                            <Double>0.0</Double>          <!--    H2S -->
                            <Double>0.459217</Double>          <!--    Formaldehyde -->
                            <Double>0.0</Double>          <!--    HCN -->
                            <Double>0.100472</Double>          <!--    Ethylene -->
                            <Double>0.528250</Double>          <!--    N2 -->
                            <Double>0.429766</Double>          <!--    N2O -->
                            <Double>0.016636</Double>          <!--    NH3 -->
                            <Undefined/>                  <!--    NO -->
                            <Double>1.89844</Double>          <!--    NO2 -->
                            <Double>0.097192</Double>          <!--    O2 -->
                            <Double>0.0</Double>          <!--    SO2 -->
                            <Double>0.268586</Double>          <!--    TotalSulphur -->
                            <Double>13.71810</Double>          <!--    TotalVOC -->
                            <Double>0.0</Double>          <!--    mXylene -->
                            <Double>0.0</Double>          <!--    oXylene -->
                            <Double>0.228210</Double>          <!--    pXylene" -->
                        </Data>
                    </AggregatedDataValue>
                </Data>

                </Object>
            </Components>
        </Object>
    </Components>
```

Grid | Schema | WSDL | XBRL | Authentic | Browser